

Week 3 – §3.3–3.5: Inverse, deelruimten en de oplossingsverzameling

Studiegids uit de HC-aantekeningen (dictaat-nummers ter referentie)

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Lecture framing: “Vandaag zijn alle matrices vierkant” for the first part: $Ax = b$ with $A \in M_{n,n}$, $x, b \in \mathbb{R}^n$. The second part (§3.4, §3.5) is again for general $A \in M_{m,n}$.

1. Conclusies van vorige week (square case)

Picture after **Gauss-eliminatie** of an $n \times n$ system, two possible outcomes:

Rood — n pivots:

$$\left(\begin{array}{cccc|c} 1 & * & \cdots & * & b'_1 \\ 0 & 1 & \cdots & * & b'_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & b'_n \end{array} \right)$$

Work from bottom to top: $x_n = b'_n$, $x_{n-1} = *x_n + b'_{n-1}$, $\dots$, $x_1 = *x_2 + \dots + *x_n + b'_1$. **All values are uniquely fixed.**

Blauw — **fewer than n pivots**, e.g. last row all zeros:

$$\left(\begin{array}{cccc|c} 1 & * & \cdots & * & b'_1 \\ 0 & 1 & \cdots & * & b'_2 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 0 & b'_n \end{array} \right) \quad 0 \cdot x_n = b'_n : \begin{cases} \text{strijdig} & \text{if } b'_n \neq 0, \\ x_n \text{ vrij te kiezen} & \text{if } b'_n = 0. \end{cases}$$

Bottom to top: if x_i is a pivot variable then $x_i = *x_{i+1} + \dots + *x_n + b'_i$ is fixed once $x_{i+1}, \dots, x_n$ are chosen.

Groen: #vrije variabelen = $n - \# \text{pivots}$.

Conclusie 1

For $A \in M_{n,n}$ the following are **equivalent**:

1. A has n pivot variables (after row reduction).
2. For **every** $\mathbf{b} \in \mathbb{R}^n$ the system $A\mathbf{x} = \mathbf{b}$ has a **unique** solution.
3. There **exists** a $\mathbf{b} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{b}$ has a unique solution.

($1 \Rightarrow 2$ is the red picture; $2 \Rightarrow 3$ is trivial; $3 \Rightarrow 1$ because with fewer than n pivots you never get uniqueness: either strijdig or a free variable.) **Stelling 3.3.1** in the dictaat summarises this: $r = n$ pivots $\Rightarrow$ exactly one solution; $r < n \Rightarrow$ none or infinitely many; and uniqueness for some $\mathbf{b}$ forces $r = n$.

2. Inverteerbare matrices (§3.3)

2.1 Motivation: the case $n = 1$

$ax = c$ with $a, c \in \mathbb{R}$.

- **Geval 1:** $a \neq 0$. Then $x = \frac{1}{a} \cdot c = \frac{c}{a}$ is the **unique** solution. (Picture: a non-horizontal line hits every horizontal level c exactly once.)
- **Geval 2:** $a = 0$, so $0 \cdot x = c$:
 - 2a: $c = 0 \rightarrow 0x = 0$: **infinitely many** solutions.
 - 2b: $c \neq 0 \rightarrow$ **no** solution (strijdig).

Vandaag: how do we generalise “ $a \neq 0$, so $1/a$ exists” to $n > 1$?

2.2 Identity matrix and the definition

Notatie. For $n \geq 1$, I_n is the $n \times n$ **identiteitsmatrix**:

$$(I_n)_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}, \quad I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

For every $A \in M_{n,n}$: $IA = AI = A$.

Def. 3.3.3. Let $A \in M_{n,n}$.

1. A matrix $B \in M_{n,n}$ is called **the inverse of A** if $AB = I_n$. Notation $B = A^{-1}$ (**not** $1/A$).
2. A is **inverteerbaar** if A has an inverse.

Opmerkingen (both are part of Stelling 3.3.4, proved below): 1. If $AB = I_n$ then automatically $BA = I_n$ too. 2. If A has an inverse, it is **unique**.

2.3 Two small examples (slide)

VB1. Does $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ have an inverse? Try

$$\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a + 2c & b + 2d \\ 2a + 4c & 2b + 4d \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

We would need $a + 2c = 1$ **and** $2a + 4c = 0$, i.e. $2(a + 2c) = 0$: contradiction. **No inverse.**

VB2. $\begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 5 & -2 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, so this one **is** invertible.

2.4 Stelling 3.3.4 (combined with Conclusie 1)

Let $A \in M_{n,n}$. The following are equivalent:

1. A is inverteerbaar.
2. A has n pivots after row reduction.
3. For every $\mathbf{c} \in \mathbb{R}^n$, $A\mathbf{x} = \mathbf{c}$ has a unique solution.
4. There is a $\mathbf{c} \in \mathbb{R}^n$ such that $A\mathbf{x} = \mathbf{c}$ has a unique solution.

(2 $\iff$ 3 $\iff$ 4 is Conclusie 1. So the new content is 1 $\iff$ 3.)

Proof of 3 $\Rightarrow$ 1 (“how do we build B with $AB = I$?”)

Algemene opmerking (very useful on its own): if B has columns $\mathbf{b}_1, \dots, \mathbf{b}_n$, then

$$AB = A(\mathbf{b}_1 \ \mathbf{b}_2 \ \dots \ \mathbf{b}_n) = (A\mathbf{b}_1 \ A\mathbf{b}_2 \ \dots \ A\mathbf{b}_n),$$

i.e. the i -th column of AB is A times the i -th column of B .

So $AB = I$ means

$$(A\mathbf{b}_1 \ \dots \ A\mathbf{b}_n) = (\mathbf{e}_1 \ \dots \ \mathbf{e}_n),$$

where $\mathbf{e}_i$ are the **standaardbasisvectoren** of $\mathbb{R}^n$ (the columns of I). To find $B = A^{-1}$ we must therefore solve n **systems**, each with n unknowns and n equations:

$$\boxed{A\mathbf{b}_1 = \mathbf{e}_1, \quad \dots, \quad A\mathbf{b}_n = \mathbf{e}_n.}$$

By assumption 3 each of these has a **unique** solution $\mathbf{b}_i$. Put them side by side: $B = (\mathbf{b}_1 \ \dots \ \mathbf{b}_n)$ is the unique matrix with $AB = I$. This proves 3 $\Rightarrow$ 1 **and** Opmerking 2 (uniqueness). $\square$

Proof of 1 $\Rightarrow$ 3

Assume A is invertible. Look at $A\mathbf{x} = \mathbf{c}$.

1. $\mathbf{x} = A^{-1}\mathbf{c}$ is a solution: $A(A^{-1}\mathbf{c}) = (AA^{-1})\mathbf{c} = I\mathbf{c} = \mathbf{c}$.
2. It is the **only** one: if $A\mathbf{x} = \mathbf{c}$, multiply on the left by A^{-1} : $A^{-1}A\mathbf{x} = A^{-1}\mathbf{c}$, so $I\mathbf{x} = A^{-1}\mathbf{c}$, so $\mathbf{x} = A^{-1}\mathbf{c}$. $\square$

(This step uses $A^{-1}A = I$, i.e. Opmerking 1.)

Proof of Opmerking 1: $AB = I \Rightarrow BA = I$

Same trick as 3 $\Rightarrow$ 1, but entered through statement 4.

- **Stap 1:** $B\mathbf{x} = \mathbf{0}$ has a unique solution. *Proof:* $B\mathbf{0} = \mathbf{0}$, so there is a solution. If $B\mathbf{x} = \mathbf{0}$ then $AB\mathbf{x} = A\mathbf{0} = \mathbf{0}$, and $AB = I$, so $\mathbf{x} = \mathbf{0}$. Unique.
- **Stap 2:** $B\mathbf{x} = \mathbf{c}$ has a unique solution for every $\mathbf{c}$. *Proof:* Conclusie 1, 4 $\Rightarrow$ 3, applied to B (Stap 1 is statement 4 with $\mathbf{c} = \mathbf{0}$).
- **Stap 3:** there is a matrix C with $BC = I$. *Proof:* the proof of 3 $\Rightarrow$ 1, applied to B .
- **Stap 4:** $C = A$. *Proof:* from $BC = I$: $ABC = AI = A$. But $AB = I$, so $ABC = IC = C$. Hence $C = A$, i.e. $BA = I$. $\square$

2.5 How to compute A^{-1} in practice

Solve $A\mathbf{x} = \mathbf{e}_1, \dots, A\mathbf{x} = \mathbf{e}_n$ **simultaneously** with Gauss–Jordan on $(A \mid I)$. When the left block has become I , the right block is A^{-1} .

VB. $A = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}.$

$$\begin{aligned} \left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 3 & 2 & 0 & 1 \end{array} \right) &\xrightarrow{r_2 \rightarrow r_2 - \frac{3}{2}r_1} \left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 0 & \frac{1}{2} & -\frac{3}{2} & 1 \end{array} \right) \xrightarrow{r_2 \rightarrow 2r_2} \left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 0 & 1 & -3 & 2 \end{array} \right) \\ &\xrightarrow{r_1 \rightarrow r_1 - r_2} \left(\begin{array}{cc|cc} 2 & 0 & 4 & -2 \\ 0 & 1 & -3 & 2 \end{array} \right) \xrightarrow{r_1 \rightarrow r_1/2} \left(\begin{array}{cc|cc} 1 & 0 & 2 & -1 \\ 0 & 1 & -3 & 2 \end{array} \right) \end{aligned}$$

So $A^{-1} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$. Check: $\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} 4-3 & -2+2 \\ 6-6 & -3+4 \end{pmatrix} = I$. ✓

If during the reduction the left block gets a zero row, A has fewer than n pivots and **no inverse** (VB1 above would show this).

3. Lineaire deelruimten in $\mathbb{R}^n$ (§3.4)

Def. 3.4.1. A subset $W \subseteq \mathbb{R}^n$ is a **lineaire deelruimte** of $\mathbb{R}^n$ if

1. $\mathbf{x}, \mathbf{y} \in W \Rightarrow \mathbf{x} + \mathbf{y} \in W$ (W is **gesloten onder optelling**),
2. $\mathbf{x} \in W, \lambda \in \mathbb{R} \Rightarrow \lambda \mathbf{x} \in W$ (**gesloten onder scalaire vermenigvuldiging**),
3. $\mathbf{0} \in W$.

Lecture remarks on the slide:

- Condition 3 is automatic as soon as $W \neq \emptyset$ (take $\lambda = 0$ in condition 2). Its job is to exclude the empty set.
- **Pictures:** a line *not* through $\mathbf{0}$: **not** a subspace (fails 3). A shaded region/half-plane: **not** (fails 2, scaling leaves it). A line or plane **through** $\mathbf{0}$: **yes**.
- **How to prove the properties** — example: $W = \{\mathbf{x} \in \mathbb{R}^2 \mid A\mathbf{x} = \mathbf{0}\}$. Property 1: let $\mathbf{x}, \mathbf{y} \in W$ be arbitrary. Then $A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = \mathbf{0} + \mathbf{0} = \mathbf{0}$, so $\mathbf{x} + \mathbf{y} \in W$. (Properties 2 and 3 go the same way: $A(\lambda \mathbf{x}) = \lambda A\mathbf{x} = \mathbf{0}$ and $A\mathbf{0} = \mathbf{0}$.) This set is the **nulruimte** $\text{Nul}(A)$, Def. 3.4.7 / Voorbeeld 3.4.8.

Opansel / Span (Def. 3.4.3)

For vectors $\mathbf{v}_1, \dots, \mathbf{v}_r \in \mathbb{R}^n$:

$$\text{Span}(\mathbf{v}_1, \dots, \mathbf{v}_r) = \{\lambda_1 \mathbf{v}_1 + \dots + \lambda_r \mathbf{v}_r \mid \lambda_1, \dots, \lambda_r \in \mathbb{R}\}$$

= the set of all **lineaire combinaties** of $\mathbf{v}_1, \dots, \mathbf{v}_r$. A span is always a linear subspace (Stelling 3.4.4).

VB. All three of these are the **grondvlak** (x_1x_2 -plane) in $\mathbb{R}^3$:

$$\text{Span}\left(\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right) = \text{Span}\left(\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix}\right) = \text{Span}\left(\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}\right).$$

Any two *onafhankelijke* vectors in the plane span it; the third vector in the last span is **overbodig** (redundant) “maar mag wel”. Week 4 makes “redundant” precise.

4. De meetkunde van de oplossingsverzameling van $A\mathbf{x} = \mathbf{c}$ (§3.5)

Now $A \in M_{m,n}$, $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{c} \in \mathbb{R}^m$.

Def. 3.5.1. - $\mathbf{c} = \mathbf{0}$: the system is **homogeneous**; solution set $S_{\text{hom}} = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\}$ ($= \text{Nul}(A)$).
 - $\mathbf{c} \neq \mathbf{0}$: the system is **inhomogeneous**; solution set $S_{\text{inhom}} = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{c}\}$.

Opmerking (bewijs zelf): 1. S_{hom} is a lineaire deelruimte of $\mathbb{R}^n$ (it is exactly the W proved above). 2. S_{inhom} is **not** a lineaire deelruimte ($\mathbf{0} \notin S_{\text{inhom}}$ since $A\mathbf{0} = \mathbf{0} \neq \mathbf{c}$).

Stelling 3.5.3

Let $\mathbf{x}_0 \in S_{\text{inhom}}$ be one fixed solution (**any** solution will do — compare with the steunvector of a line/plane). Then

$$S_{\text{inhom}} = \{\mathbf{x}_0 + \mathbf{y} \mid \mathbf{y} \in S_{\text{hom}}\}$$

Picture: S_{hom} is a line/plane **through 0**; S_{inhom} is the same line/plane **shifted** so that it passes through $\mathbf{x}_0$. This is exactly what you saw in week 2 VB2/VB3: steunvector + span of direction vectors.

Bewijs (set equality = two inclusions). Fix $\mathbf{x}_0 \in S_{\text{inhom}}$.

- $\{\mathbf{x}_0 + \mathbf{y} \mid \mathbf{y} \in S_{\text{hom}}\} \subseteq S_{\text{inhom}}$. Let $\mathbf{y} \in S_{\text{hom}}$ be arbitrary. Then $A(\mathbf{x}_0 + \mathbf{y}) = A\mathbf{x}_0 + A\mathbf{y} = \mathbf{c} + \mathbf{0} = \mathbf{c}$, so $\mathbf{x}_0 + \mathbf{y} \in S_{\text{inhom}}$.
- $S_{\text{inhom}} \subseteq \{\mathbf{x}_0 + \mathbf{y} \mid \mathbf{y} \in S_{\text{hom}}\}$. Let $\mathbf{z} \in S_{\text{inhom}}$ be arbitrary. We show there is a $\mathbf{y} \in S_{\text{hom}}$ with $\mathbf{z} = \mathbf{x}_0 + \mathbf{y}$. Look at $A(\mathbf{z} - \mathbf{x}_0) = A\mathbf{z} - A\mathbf{x}_0 = \mathbf{c} - \mathbf{c} = \mathbf{0}$. So $\mathbf{z} - \mathbf{x}_0 \in S_{\text{hom}}$; call it $\mathbf{y}$. Then $\mathbf{z} = \mathbf{x}_0 + \mathbf{y}$ with $\mathbf{y} \in S_{\text{hom}}$. $\square$

Practical consequence. To describe all solutions of $A\mathbf{x} = \mathbf{c}$ you need (i) one particular solution $\mathbf{x}_0$ and (ii) the solutions of the homogeneous system. Gauss–Jordan on $(A \mid \mathbf{c})$ gives both at once: the constant vector is $\mathbf{x}_0$, the vectors multiplied by free parameters span S_{hom} .

Zelftest

- $A \in M_{4,4}$ reduces to a matrix with 3 pivots. How many solutions can $A\mathbf{x} = \mathbf{b}$ have? Is A invertible? *Answer:* either none or infinitely many (never exactly one). Not invertible (needs 4 pivots).
- Why is “ $B = 1/A$ ” bad notation? *Answer:* there is no division of matrices; A^{-1} is defined by $AB = I$, and since $AB \neq BA$ in general one has to *prove* that $BA = I$ as well (Opmerking 1).
- Find the inverse of $\begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}$ with $(A \mid I)$. *Answer:* $r_2 \rightarrow r_2 - 2r_1$: $\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right)$;
 $r_1 \rightarrow r_1 - 2r_2$: $\left(\begin{array}{cc|cc} 1 & 0 & 5 & -2 \\ 0 & 1 & -2 & 1 \end{array} \right)$. So $A^{-1} = \begin{pmatrix} 5 & -2 \\ -2 & 1 \end{pmatrix}$, matching VB2.
- Is $\{(x_1, x_2)^t \mid x_1 + x_2 = 1\}$ a subspace of $\mathbb{R}^2$? *Answer:* no, $\mathbf{0}$ is not in it. (It is an S_{inhom} .) Is $\{(x_1, x_2)^t \mid x_1 + x_2 = 0\}$? *Answer:* yes (an S_{hom}).
- In week 2 VB3 the solution set was $\mathbf{p} + s\mathbf{a} + t\mathbf{b} + u\mathbf{c}$. Identify $\mathbf{x}_0$ and S_{hom} . *Answer:* $\mathbf{x}_0 = (1, 0, -1, 0, 0)^t$, $S_{\text{hom}} = \text{Span}(\mathbf{a}, \mathbf{b}, \mathbf{c})$ with $\mathbf{a} = (-2, 1, 0, 0, 0)^t$, $\mathbf{b} = (-1, 0, -2, 1, 0)^t$, $\mathbf{c} = (-1, 0, -1, 0, 1)^t$.